The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Computer Networks

Lab No# 02

Task Title: Learning Networking Basics & Creating LAN

Submitted to: Engr.Dr.Asma Javed

Student Name: Feeza Kulsoom

Roll No: 2024-SE-03

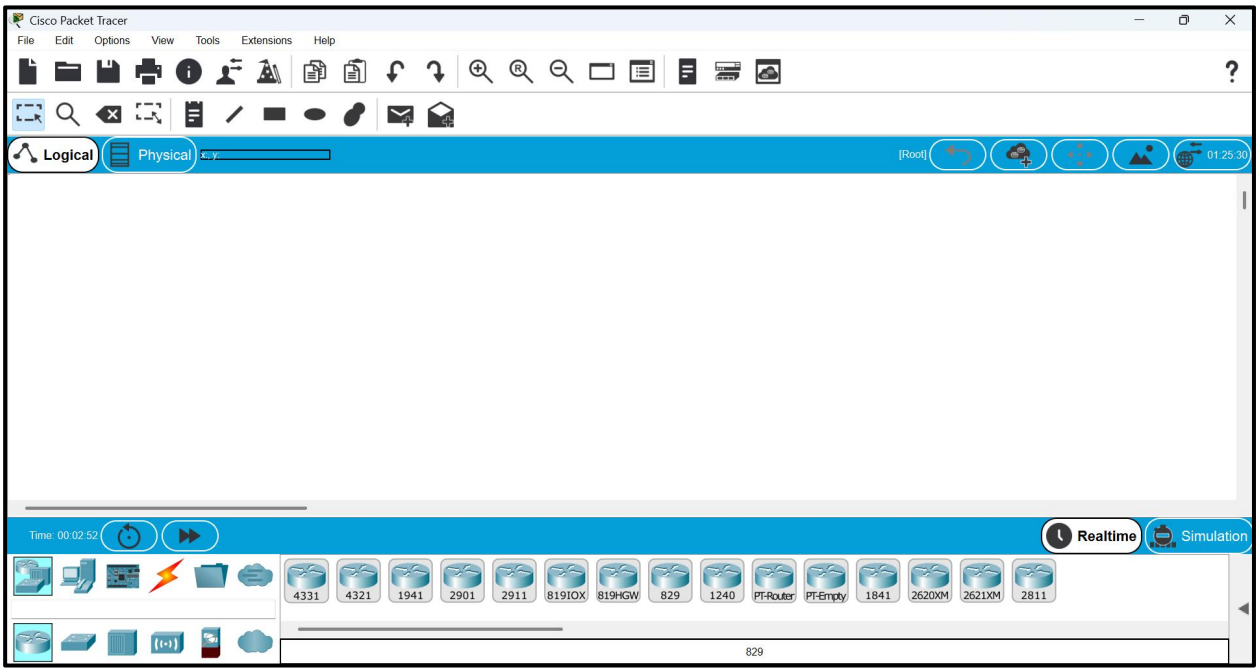
Reg Number: 2024-UMDB-004738

Objective

The objective of this lab is to understand the basic concept of computer networks, identify common networking devices, explore Cisco Packet Tracer, and create a simple Local Area Network (LAN). The lab also introduces basic networking commands used to test connectivity between devices.

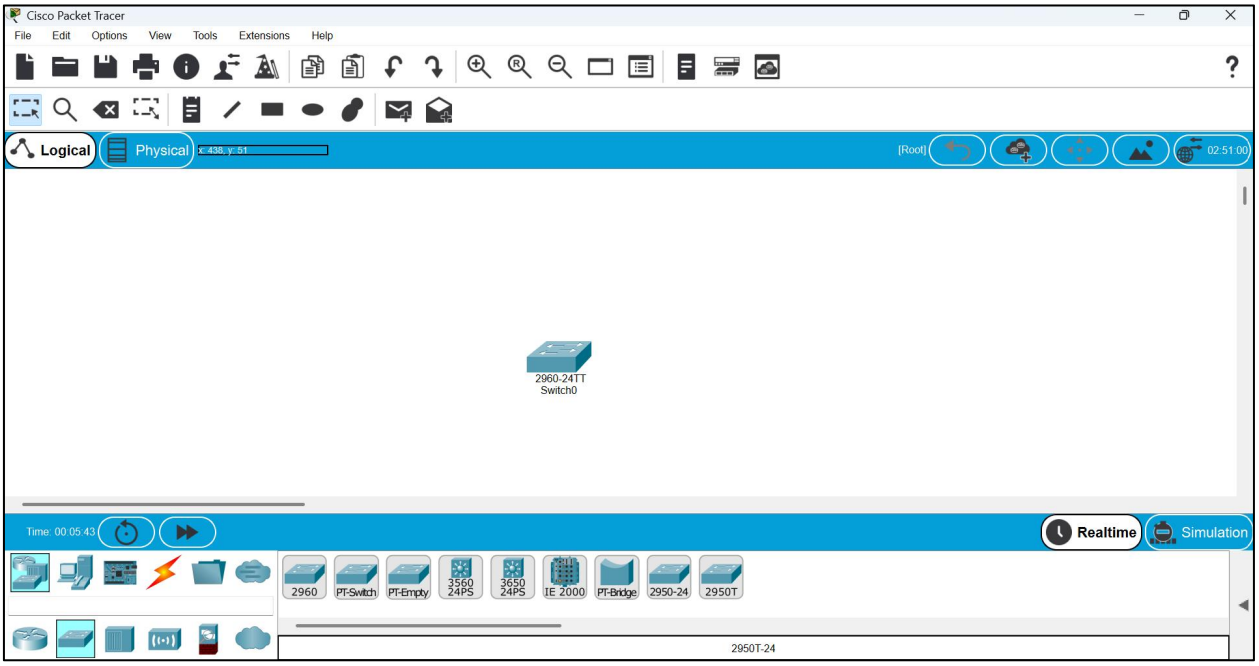
■ Step 1: Opening Cisco Packet Tracer

Cisco Packet Tracer is a network simulation tool used to design and test network topologies. First, launch Cisco Packet Tracer from your computer. When the application opens, the main interface will appear showing the workspace, device panel, and connection tools.



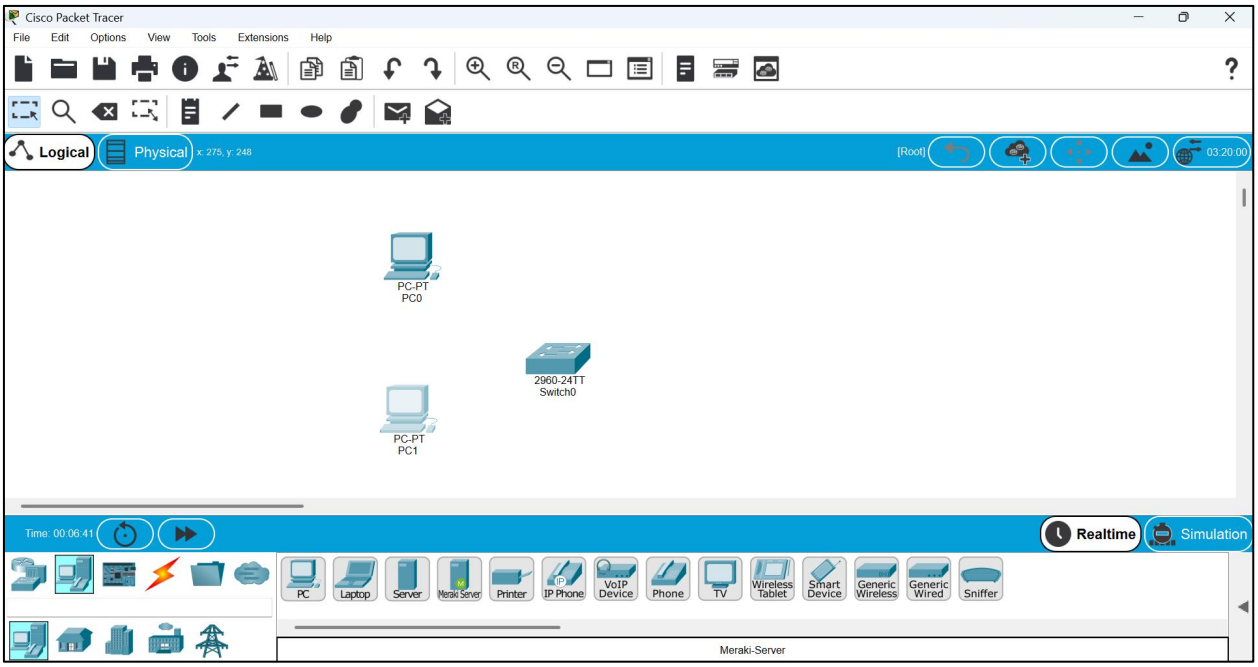
■ Step 2: Adding a Switch to the Workspace

A switch is a networking device used to connect multiple devices within the same network. From the 'Network Devices' section, select 'Switches' and drag a 2960 switch into the workspace.



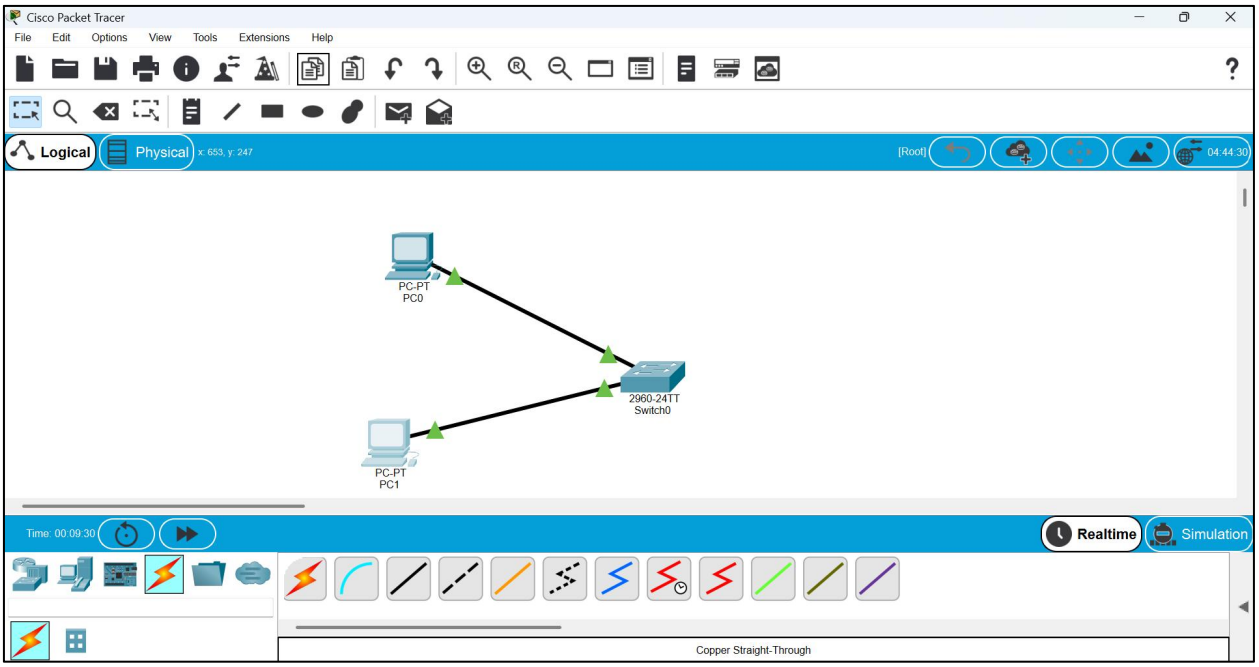
■ Step 3: Adding End Devices (PCs)

End devices represent computers or hosts in the network. From the 'End Devices' section, drag two PCs into the workspace. These devices will be used to test communication.



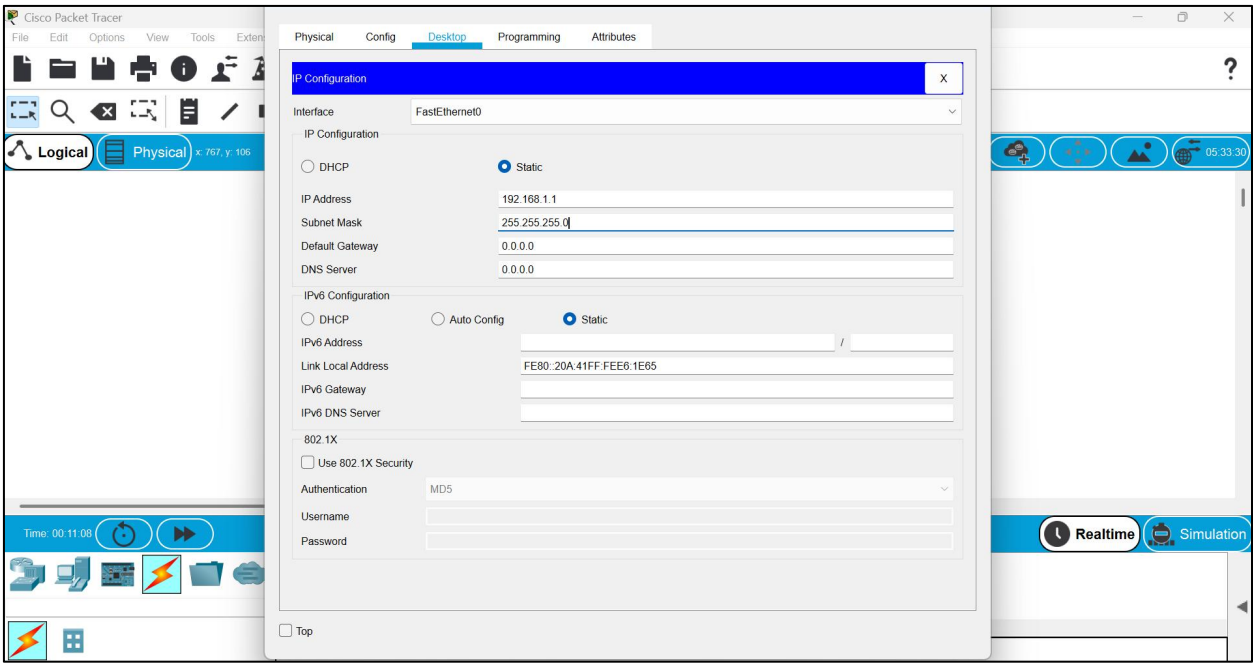
■ Step 4: Connecting Devices Using Cables

To allow communication, devices must be connected using network cables. Select 'Copper Straight-Through' cable from the connections panel. Connect PC0 to the switch and then connect PC1 to the switch.



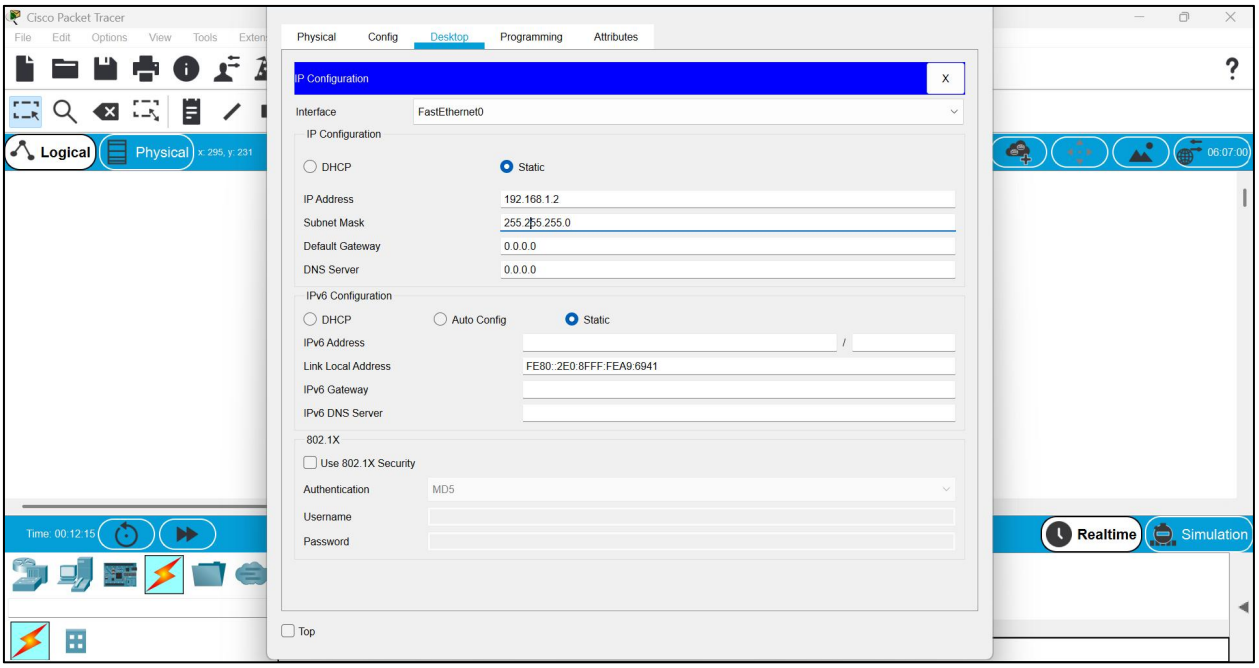
■ Step 5: Configuring IP Address for PC0

Every device in a network must have a unique IP address. Click PC0, go to the Desktop tab, open IP Configuration, select Static, and assign IP Address 192.168.1.1 with Subnet Mask 255.255.255.0.



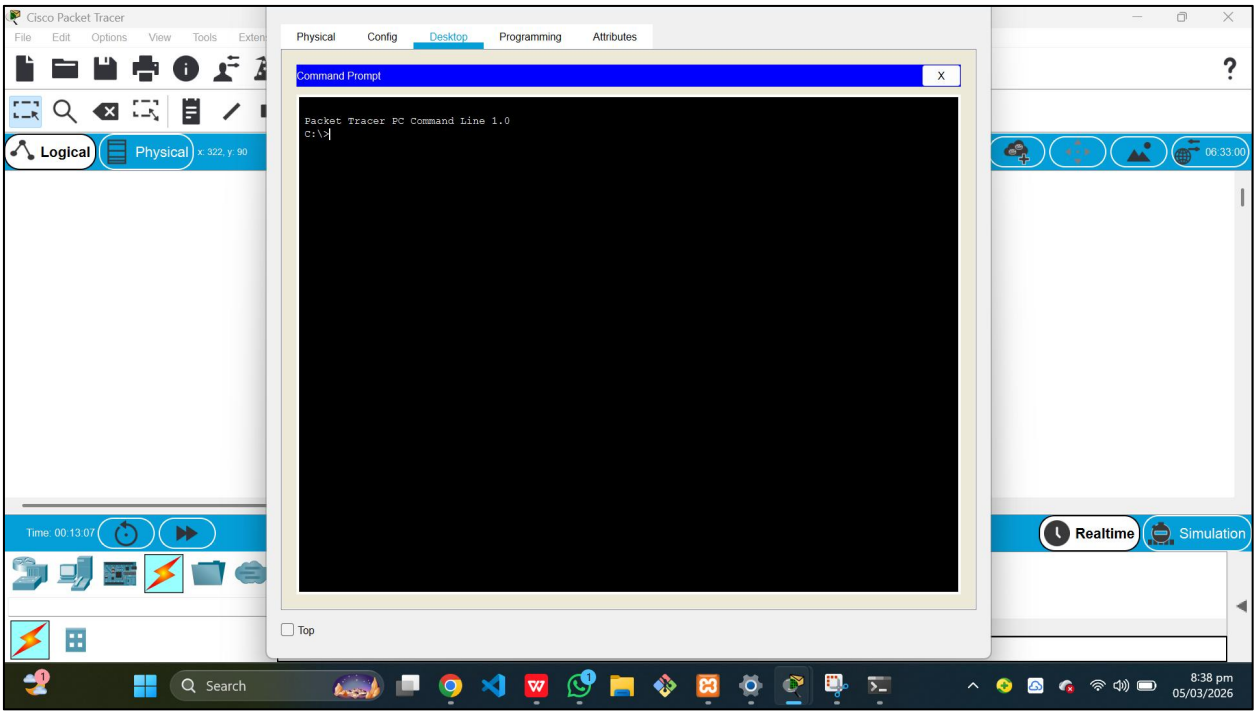
■ Step 6: Configuring IP Address for PC1

Repeat the same process for PC1. Open Desktop → IP Configuration → Static and assign IP Address 192.168.1.2 with Subnet Mask 255.255.255.0. Both PCs must be in the same network to communicate.



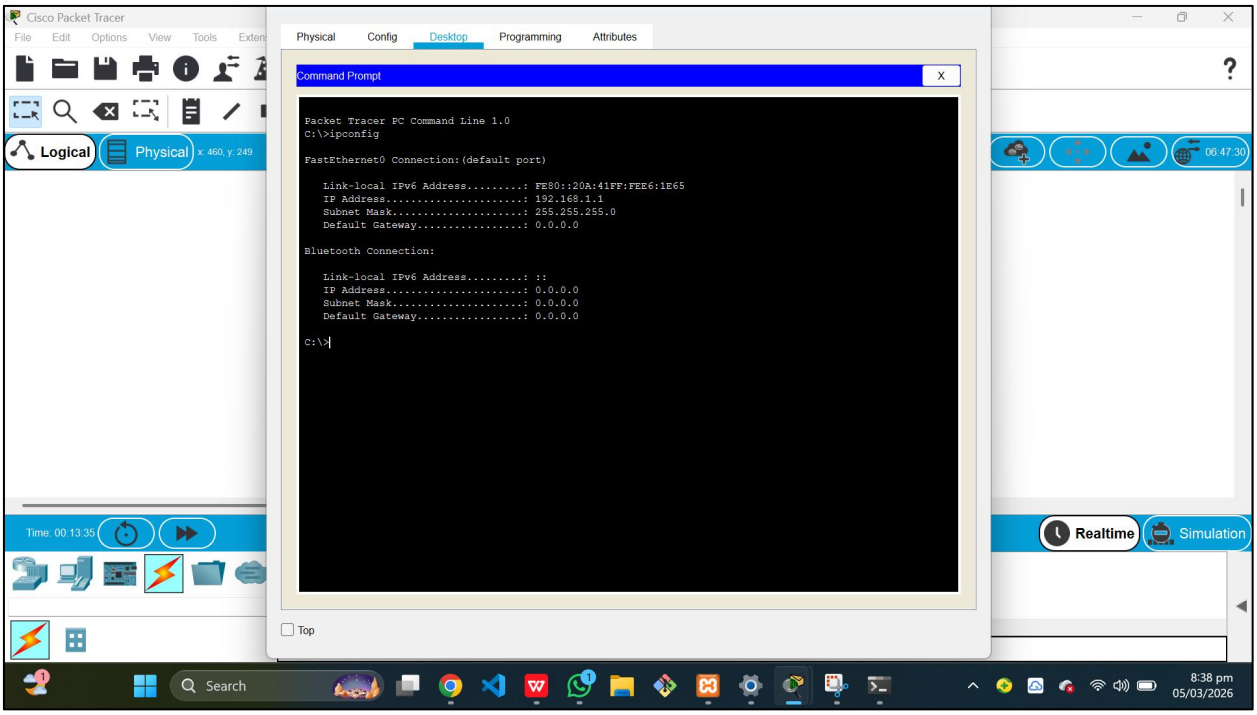
■ Step 7: Opening Command Prompt

To test the network, open the Command Prompt. Click on PC0, go to the Desktop tab, and select Command Prompt.



■ Step 8: Running the ipconfig Command

The ipconfig command displays the current IP configuration of the device including the IP address and subnet mask.



■ Step 9: Testing Connectivity Using ping

The ping command checks whether two devices in the network can communicate. From PC0, type 'ping 192.168.1.2'. If the network is correctly configured, you will receive reply messages.

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

■ Step 10: Running the tracert Command

The tracert command shows the path taken by packets to reach the destination device. In a simple LAN with a switch, the output usually shows a single hop.

```
C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1   2 ms      5 ms      1 ms      192.168.1.2

Trace complete.

C:\>
```

Top

■ Step 11: Viewing ARP Table

The arp -a command displays the ARP table which maps IP addresses to MAC addresses of devices in the network.

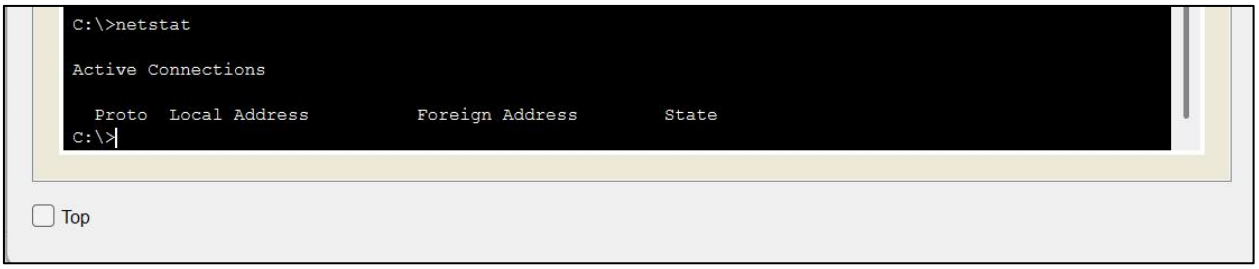
```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.2           00e0.8fa9.6941        dynamic

C:\>
```

Top

■ Step 12: Running netstat Command

The netstat command displays network statistics and active connections. This helps in monitoring network activity.



■ Conclusion

In this lab, a simple LAN network was created using Cisco Packet Tracer. Two PCs were connected through a switch and configured with IP addresses. Different networking commands such as ipconfig, ping, tracert, arp, and netstat were used to test connectivity and analyze network communication.